Cloud Forensics Challenges in Bangladesh

A PRISMA Systematic Review with AI-Enhanced Resolution Strategies

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PRISMA SLR | Cloud Forensics BD | AI-CFRF-BD | Section 65B | MLAT

Motivation: Why Does Bangladesh Have a Problem?

100M+

internet-enabled
devices in BD 2023

$1.2B

public cloud procurement
2020–2024

3 Laws

ICT 2006 · DSA 2018 · CSA 2023
All silent on cloud evidence

0

statutory data-hold
mechanisms in BD law

Why Cyber Tribunal Cases Collapse

Log Expiry (30–90 days)

CSP logs auto-deleted before any court order can be obtained. Bangladesh has no statutory mechanism to pause deletion.

Section 65B Failure

Requires physical device custody — structurally impossible for remotely hosted cloud logs. Precondition can never be met.

Broken Chain-of-Custody

No contemporaneous record of CSP-to-court log transfer. Forensic reports ignore the cloud storage layer entirely.

MLAT Latency: 6 wk–10 mo

Bangladesh processes fewer than 50 MLAT requests/year. No Budapest Convention. No bilateral CSP data-sharing.

Research Questions & Contributions

RQ1	What technical challenges do cloud environments pose to digital forensics in Bangladesh?
RQ2	What legal barriers prevent admissibility of cloud-sourced evidence under Bangladesh's legal framework?
RQ3	How do existing AI-enhanced tools address the challenges in RQ1 and RQ2?
RQ4	What implementable framework can improve cloud forensic readiness for BD law enforcement?

Three Contributions

① **First PRISMA SLR with BD legal context**

16 open-access studies · 2021–2026 · Table I

② **Five-Cluster Taxonomy C1–C5**

Each cross-referenced against BD legal gaps · Table II

③ **AI-CFRF-BD Framework**

5 validated AI tools · deployable within existing BD law

Methodology (PRISMA) & Five-Cluster Taxonomy — Table II

PRISMA FLOW

n = 782
7 databases searched

n = 534
248 duplicates removed

n = 48
486 excluded off-topic

n = 16
32 excluded (18 paywall + 14 peripheral)

Thomas & Harden synthesis · Single reviewer (limitation) · ISO/IEC 27037

CHALLENGE TAXONOMY (TABLE II)

C1	**Volatile Data & Log Expiry**	CIAF — LLM	93% F1
C2	**Multi-Tenancy & Evidence Contamination**	ML-CF — RF/LSTM/CNN	ISO 27043
C3	**Chain-of-Custody & Section 65B**	Blockchain CoC	14/17 jurisd.
C4	**Cross-Border MLAT Latency**	CLERMS — NLP	40–70% ↓
C5	**Forensic Readiness Deficit**	Self-Hosted CF Platform	60% cost ↓

All 5 tools deployable within BD's existing legal authority — no new legislation required.

Limitations

L1

Small Corpus — Qualitative Only

16 studies: complete open-access coverage within scope but precludes quantitative meta-analysis. All findings are qualitatively synthesised.

L2

Single Reviewer — No Inter-Rater Reliability

All thematic coding performed by the author alone. No second coder verified cluster assignments. Acknowledged as a methodological limitation.

L3

Legal Currency — 2025 Instruments Evolving

Legal analysis current as of June 2026. Secondary regulations under PDPO 2025 and CSO 2025 are yet to be enacted and may alter the landscape.

L4

No BD Empirical Validation of AI Tools

All five AI-CFRF-BD tools were validated in US/EU environments. Performance within Bangladesh's regulatory and technical context is untested — primary future work direction.

Research Gap — What Table I Shows

Ref	Focus & Contribution	C1	C2	C3	C4	C5	AI Tech.	BD Rel.	Method	Year
[1]	Alshabibi et al. — SLR, 7 tech + 5 legal barriers	✓	✓	■	■	✓	ML survey	✗	SLR	2024
[5]	Alharthi & Garcia — CIAF LLM (93% F1)	✓	✓	■	✗	■	LLM/GenAI	■	Framework	2025
[6]	Igonor et al. — Blockchain CoC, 14/17 jurisd.	✗	■	✓	✓	✗	Blockchain	■	Prototype	2025
[7]	Tageldin & Venter — ML-CF, ISO/IEC 27043	✓	✓	■	✗	✓	ML/LSTM	■	SLR	2023
[9]	Akilal & Kechadi — CLERMS, 40–70% MLAT	✓	✗	✗	✓	■	NLP/Log	■	Prototype	2025
[12]	Ahmed & Arifuzzaman — CSA 2023 BD gap	✓	✓	✓	■	✓	Statistical	✓	Survey	2025
[13]	Akter & Rahman — cloud forensic dims., UIU	✓	✓	✓	✓	■	Process ML	✓	SLR+Model	2023
[15]	Sharmin & Rahman — BD Police survey (52)	■	✗	✓	✓	✓	Statistical	✓	Survey	2025
[P]	**This study — AI-CFRF-BD (PRISMA 2020 SLR)**	✓	✓	✓	✓	✓	**LLM+ML+BC**	✓	**PRISMA**	**2026**

Gap: All AI-tool studies [5–7,9] validated in US/EU only — none engages with Section 65B or BD data-hold absence. BD survey studies [10,11,15] document failures but provide no AI pathway. [P] is the only work covering all 5 dimensions with BD relevance.

Summary

✅ Delivered (Sections I–III)

- PRISMA SLR — 16 studies, 2021–2026
- Problem: 4 Tribunal failure modes mapped
- Five-cluster taxonomy C1–C5 (Table II)
- Comparative analysis: 8 prior studies (Table I)
- AI-CFRF-BD: 5 tools, all within existing BD law
- Four limitations explicitly stated

➡️ Full Submission Adds

IV **Challenges in BD**

C1–C5 analysis → RQ1 & RQ2

V **AI-CFRF-BD Detail**

5 cluster–tool pairings → RQ3 & RQ4

VI **Full Limitations**

Scope, legal currency, future work

Thank you · Questions welcome

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